ARTICLE

The First Fifteen Years of Computing in Aotearoa New Zealand

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Abstract—This article outlines the first fifteen years of modern computing in New Zealand ...

Modern computers arrived in Aotearoa New Zealand in 1960, less than a century after the first discernable information technology [25]. New Zealand was then a relatively isolated and small (2.5 million people) economy, so it is feasible to track the dissemination and socio-economic influence of computing for the following period; this study runs from 1960 to about 1975. Two events in the mid-1970s signaled the end of the economic and social era that began after World War II, and act as the end of our study: Britain joined the European Common Market in 1973, with major economic consequences, and New Zealand switched on a national Police Computer in 1976, with significant social impact.

In 1960, New Zealand had a rather centrally managed economy. Indeed, its currency did not float until as late as 1985. The details are complex [38], but the result was that during the period of our study the Treasury was always concerned about the foreign exchange impact of computer imports, and this necessitated a system of import licensing. This was a constant background for trends in computing, especially since the Treasury initially favored computing service bureaus to ensure maximum usage of what they considered a scarce resource. Of course, many companies much preferred to have their own systems.

Several themes are used to organize this article:

- growth in numbers of computers, vendors and employment
- key areas and types of usage, including service bureaus
- start of computing services and teaching in universities, technical colleges
- professionalization
- local contributions to the technology
- commercial and economic impact
- social impact

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GROWTH IN NUMBERS AND EMPLOYMENT

The following table shows the estimated number of electronic digital computers installed in New Zealand over the years in question.

Year	Computers	Source	Notes
1960	2	[24]	
1965	70	[21]	[30] gives 45
1966	81	[21]	
1967	100	[3]	
1968	120	[21]	[12] gives 87
1969	140	[21]	
1971	180	[21]	
1972	200	[12]	
1974	280	[21]	
1976	400	[21]	

A snapshot of installations in Auckland alone lists 37 sites at the end of 1969, covering businesses of many sorts, local government, a university, and even the Naval Research Laboratory [20]. At least seven of the sites appear to be service bureaus. The author noted in passing that there was also a large Government investment in computers in Wellington.

Although there are some discrepancies in the available data, the trend is one of rapid growth, from one computer for every 1.2 million people to one for every 8000. Similarly, the number of programmers in public services rose from a handful (maybe 4) in 1960 to 115 (84 men and 31 women) by 1974 [12]. Overall, 4000 data processing staff were then employed at 220 sites – 39% in Wellington, 31% in Auckland, 9% elsewhere in the North Island, and 21% in the South Island [21]. Pay was attractive: in 1967, an EDP manager's average annual salary was $4580NZ [3], when the

FIGURE 1. Fletcher Computing Bureau staff, 1966. Courtesy Fletcher Archives, NZ.

national average for company directors and managers was $4220NZ [5]. Programmers and systems analysts earned anywhere between $2000 and $4000.

Even if the public services had 27% of female staff, it is not clear that the same applied elsewhere. For example, a 1966 staff photograph of Fletcher Building's internal computer bureau shows 14 men and only three women (Figure 1). Whenever staff appear in computer room photographs, the women are almost invariably seated at a keyboard and the men are standing in discussion. A 1967 survey [3] shows systems design and programming as 90% male occupations, operations as 50% male, and data preparation as 97% female.

There is little to indicate significant Māori involvement in computing during our period of study. Robyn Kāmira [32] shows that such involvement would only start in the 1980s, when a first handful of Māori information technology specialists graduated.

New Zealand had a long history of preferring to import British products of all kinds, but this did not long survive in the computer market. IBM was of course a major player; not only did they snag the Treasury as their first customer for an IBM 650 in 1960 [24], but they also delivered a heavily discounted IBM 1620 to the University of Canterbury in 1962, the first machine in academia [16]. This was in line with IBM's world-wide practice of educational discounts, a powerful marketing tool that exposed many students to the IBM brand. As time went on, IBM sold numerous machines - more IBM 1620s, at least one IBM 1401, several IBM 1130s, until the game-changing IBM 360 series came along in 1964.

The main British competitor in 1960 was ICT, but a potential customer would certainly have been confused by the fragmented state of the British computer industry at that time [29]. Nevertheless, ICT sold one ICT 1201, at least six ICT 1301s, and numerous machines from the ICT 1900 series. It was not until the various British companies were unified as ICL in 1968 that things settled down, but by then IBM had the lion's share of the New Zealand market and ICL was falling behind. In 1974, the new ICL 2900 series had to compete with the well established and upwards compatible IBM 360 series.

Other companies that were active in the early days were Burroughs (with its E2000, not a general-purpose computer, but specialized for accounting), and NCR (Series 500, also specialized for ledgers).

We have collected available information about individual computers present in New Zealand during the period under study, but at best it is a small sample of the market. At the time of writing, we have identified 81 **[TBD: Check!]** machines (including programmable calculators) with arrival dates up to 1974, but the estimated total in that year was 280. Many of the earlier machines had of course been scrapped by then, so our sample certainly covers less than 25% of the market. The distribution of this sample across manufacturers is shown in the following table. (The total for the British companies that ended up as part of ICL is 29, similar to the total for IBM.)

Manufacturer	Count
Burroughs	12
DEC	9
EELM	3
Elliot	1
HP	1
IBM	28
ICL	8
ICT	17
NCR	2

Throughout the period under study, most computers remained large and comparatively heavy, even as they moved from "first generation" (vacuum tubes) through "second generation" (discrete transistors) and "third generation" (integrated circuits) to the "fourth generation" (very large scale integration). As the technology got smaller, mainframe computers became more powerful rather than more compact; space and air conditioning requirements did not substantially change. For example, the third computer installed in 1969 by Motor Specialties in Auckland, an ICL 1902A, needed to be lifted to the top floor by crane, like its predecessors, an ICT 1201 and an ICT 1301 (Figure 2). Only after 1970 when minicomputers such as the DEC PDP-11 entered the market did this change, so that

FIGURE 2. ICL 1902A Delivery in 1969. Courtesy Fletcher Archives, NZ.

FIGURE 3. The TAB Computer Centre, Christchurch, 1970. Courtesy Fletcher Archives, NZ.

some computers could be installed almost as easily as domestic appliances.

Originally, much equipment was rented; in 1967, 70% of “owners” actually rented their equipment, and a further 15% rented part of it [3]. By the late 1960s, the growth in installations led to a minor building boom of dedicated computer centers, which tended to be stark concrete buildings with few windows (Figure 3). A rare exception was Ceramic House in Auckland (Figure 4), specifically designed to accomodate an ICT 1901 delivered in 1969. This building was designated as a heritage site in 2024.

This section has described how computers were imported, housed, installed and staffed during the period of study. We will now consider what they were used for.

FIGURE 4. Ceramic House, 1969. Courtesy Auckland Libraries Heritage Collections.

HOW COMPUTERS WERE USED

In 1960, the main expected workload was the payrolls for teachers and public servants. The expansion in fields of use came very quickly. By the end of 1966 [10] [**TBD**: Janet check the Archives citation], Government usage included payroll, statistics, science, NZ Railways, Inland Revenue, Post Office, Social Security, Reserve Bank, Bank of New Zealand, New Zealand Broadcasting Corporation, Apple & Pear Board, and the Dairy Board. In a surviving list of import license applications between August 1965 and May 1966 [10], there are three applications from local government, three from universities, but 30 from business and primary and secondary industries. The New Zealand economy took to computers rather quickly, held back mainly by import license restrictions.

Applications of importance for business users were listed in 1967 as invoicing, payroll, general ledger, stock control, expense analysis and sales statistics [3]. All the same, mathematical, scientific and project planning applications such as PERT (Program Evaluation and Review Technique) were increasingly used. Users saw the advantages of computerization as speed, accuracy, improved reporting, improved organizational control and staff savings. Quite surprisingly, in 1967 most computers were underutilized, with 250 operating hours per month on average, and single-shift operation still being typical. Only 68% of installations charged internal users for usage: “one of the chief reasons why installations can be uneconomic”.

In 1973, the Department of Statistics provided the following summary [8]:

> The demand for computers has come from Government departments, local authorities, universities, primary producer boards, pri-

vate firms in industries such as printing, forestry, insurance, oil, food processing, electrical equipment manufacturing, building and construction, clothing, engineering, airways, banking, retailing, motor assembly, paint manufacturing, and stock and station agents.

Because of government's reluctance to issue import licenses, numerous service bureaus arose to allow resource-sharing. By 1967, when the total number of computers was about 100, at least 166 companies were using bureau services operated by computer vendors, and another 78 were using independent bureaus [3]. The following table lists some example bureaus and their approximates dates of foundation, but the list is far from complete.

Year	Name	Location	Notes
1961	Electronic Data Processing Ltd	Auckland	Closed 1963
1963	Electronic Data Systems Ltd	Wellington, later Auckland & Christchurch	50% EELM, 50% NZ Truth (until 1968)
1965	Computer Bureau Ltd	Christchurch, later nationwide	Renamed Datacom
1966	Allied Computer Processors	Dunedin	
1967	Databank Systems Ltd	multiple	Banking consortium
1968	Electronic Data Processing NZ Ltd	Wellington	
1968	NCR Bureau	Wellington	
1968	Burroughs Bureau	Wellington	
1968	ICL Centre	Auckland	
1969	Computer Activities	Auckland	
1969	Computer Systems	Auckland	
1969	Data Enterprises Ltd	Tauranga	Paul and David Walker

The first bureau listed above remains something of an enigma. According to ICT's records in the U.K., it bought an ICT 1301 for "service work"; but we have found no local records to substantiate this, beyond the formal records of the formation and dissolution of the company. Another phantom bureau hosted the government Education Department's second computer, also an ICT 1301, leased from ICT at the end of 1963 and installed in ICT premises in Wellington. When the Minister of Education pressed a button to inaugurate it, it was said to be "New Zealand's first computer bureau" [2], although all indications are that it was dedicated to the Education Department. It seems that obtaining an import license for a shared bureau computer was significantly easier than for a single-user machine. The other bureaus were real enough, and we now present a few brief case studies.

FIGURE 5. EDS KDF6, Princes St, Auckland in about 1970. Courtesy Neil Harsant.

Electronic Data Systems Ltd (EDS), founded in 1963, was jointly owned by the British company English Electric Leo Marconi (EELM) and the NZ Truth newspaper. It had no connection with the later American company of the same name. EELM was one of the numerous firms that ended up as part of ICL, and presumably wanted to enter the New Zealand market. Apart from its earlier DEUCE and LEO computers, it made a family of KDF machines, including the KDF6, of which at least 12 were sold [29]. The KDF6 sales records are scanty [17], but EDS had such machines in Wellington and Auckland [28]. Indeed we have photographic evidence of the machine in Auckland (Figure 5).

NZ Truth, described by the National Library web site as "one of the country's most colourful, controversial and popular newspapers" ran from 1905 until 2007. It is unclear why it decided to enter the computer bureau business in 1963. In any case, EDS opened a third branch in Christchurch in 1970, and in due course switched to Burroughs and ICL equipment [7]. By 1972, the Christchurch and Wellington offices were linked by a remote job entry system. EDS remained in business in one form or another until 1994, decades after EELM no longer existed and NZ Truth had sold its stake.

Another success story started in 1965. G. Bernard Battersby (1925-1997) was the Dean of the Faculty of Commerce at the University of Canterbury, and a government advisor. Yet he joined with Paul M. Hargreaves (1938-2014) to found Computer Bureau Ltd (CBL) in Christchurch [14]. Its first computer was an ICT 1902 delivered in 1966; with a staff of seven, CBL was unprofitable until 1969, when a branch was opened in Wellington. In early 1970, CBL had some 50 customers, and the Christchurch computer was working 140 hours per week. Thus an ICL 1902A was delivered to Christchurch, the orginal 1902 was moved to Wellington, and an additional ICL 1902 was ordered for another new branch in Hamilton. In Auckland, CBL had taken over Fletcher Building's in-house computer bureau, also ICL-based. All training of systems analysts and programmers took place in Christchurch, with total staff expected to reach 70 during 1970 [4]. By 1974, CBL needed to order three ICL 1902T models for Auckland, Wellington and Christchurch, with a nationwide total of six computer centers and 125 staff. Furthermore, they told the newspapers [9]:

> [CBL] is unique in that many of its major users are also shareholders, a number of whom previously operated their own inhouse computers. These include Fletcher Holdings, Ltd, Wilson and Horton, Ltd, the New Zealand Co-op Dairy Company, Bing Harris Sargood, Ltd, United Dominion Corporation, Ltd, and in Christchurch, the Canterbury Building Society, the S.I.M.U. Mutual Insurance Association, and the City Council.

Clearly, Computer Bureau Ltd had found a winning formula. In 2025, they still prosper under the name of Datacom, claimed to be the largest technology company in New Zealand, employing at least 6000 people at offices across the Asia-Pacific region (Figure 6).

TBD: short case study of Databank [Janet]

Thus, as computer usage spread throughout business and government, import license pressure encouraged resource sharing and successful bureau operations. This pressure was not to be relieved until major economic reforms in the 1980s, outside our study period.

FIGURE 6. Datacom, Auckland, 2025. Courtesy Brian Carpenter.

PROFESSIONALIZATION AND ACADEMIA

Professionalization started early. The New Zealand Computer Society was founded in the same year that the first modern computers arrived in the country...

TBD: Origin of NZCS and growth until 1975 [Janet]

Naturally, the need for training was an issue from a very early stage. By 1967, it was a serious concern: "It is not sufficient for the Universities to provide computer facilities and training only at the scientific level. The Technical and Management Institutes cannot be left to their own devices... the secondary schools (and before too long the primary schools) must introduce computer knowledge" [3]. In practice, most training originally came from computer vendors, or people learned on the job. The first NZCS national conference in 1968 led to a committee on training requirements, which recommended CS undergraduate courses at every university, as well as some postgrad and extramural courses. The committee "saw it as the primary role of Computer Science departments to educate system programmers, as well as future researchers" and recommended that "Commerce Departments should deal with teaching system analysts and managers" [15].

By 1970, technical colleges were offering courses in computing and programming, but with only 46 students enrolled nationally [8]. According to Tim Bell [22], the first complete tertiary course was offered by the Auckland Technical Institute (ATI, now Auckland University of Technology) in 1967, teaching Fortran, RPG, and COBOL, along with mathematics, statistics, and accounting. Although ATI had acquired a third-hand ICT 1201 in 1966 [24], student programs were processed overnight by a local company.

As universities acquired their own computers from 1963 onwards, they naturally started offering courses on the use of computers for mathematics, statistics, science in general, and business. Often, such courses were taught by computer center staff rather than by academic faculty. Some teaching of computing in secondary schools started from about 1974, but Bell wrote that "finding computers that students could program was a challenge, and there was no national initiative around computers in schools until the end of the 20th century." Indeed, Massey University was offering extramural computer science courses by the mid-1970s, and the most eager participants were secondary school teachers.

Among degree-granting institutions, the University of Canterbury was the first to acquire a computer, in 1962. The following table lists the first computers at each university.

Year	University	Model	Notes
1962	Canterbury, Christchurch	IBM 1620	
1963	Auckland	IBM 1620	
1965	Victoria, Wellington	Elliot 503	Shared with DSIR
1965	Massey, Palmerston North	IBM 1620	
1966	Otago, Dunedin	IBM 360/30	
1969	Waikato, Hamilton	IBM 1130	

Various other machines were purchased, culminating in a joint purchase of five Burroughs B6700 mainframes for Auckland, Massey, Wellington, Canterbury and Otago in 1971 [26]. (Waikato and Canterbury's Lincoln College campus were fobbed off with underperforming remote access to Auckland and Canterbury respectively.) Initially, these machines were primarily used in support of research in various disciplines. For example, in Auckland the IBM 1620, and an IBM 1130 that joined it in 1967, were heavily used by economists, who were "often the heaviest user," running simulations of econometric models in 1964-70. Applications, written in Fortran, included dynamic linear models of the New Zealand economy, dynamic simultaneous equation models, and systems of linear stochastic differential equations [36]. The researchers were students and colleagues of Professor Rex Bergstrom, who had cut his teeth on the Cambridge EDSAC in 1952-54. Of course, mathematicians, scientists, and engineers were also early users.

At Canterbury, the IBM 1620 was soon "heavily used by a wide range of university departments, frequently logging more than 500 hours a month... By July 1964, after 2 years of use, the University's computer had been used in more than 120 projects by 14 departments, and more than three-quarters of these projects could not have been tackled without a computer" [16].

As well as research, these early machines were also used for undergraduate programming courses. However, in the 1960s, computing was mainly taught as a useful skill, not as an academic subject in its own right. Often, it was taught by computer center staff, not by academic faculty members. The discipline of Computer Science (CS) was late to arrive in New Zealand. For example, in the UK (the largest source of academic immigrants at that time), the Mathematical Laboratory in Cambridge had offered a post-graduate Diploma in Numerical Analysis and Automatic Computing (later renamed Diploma in Computer Science) since 1953, and the Department of Computer Science at the University of Manchester was founded in 1964, graduating its first students in 1968. CS departments also emerged in the USA during the 1960s. It was not until 1973 that three CS departments were founded in New Zealand, at the University of Canterbury (Professor John Penny), Massey University (Professor Graham Tate) and Waikato University (Professor Darryl Smith) [22], [18]. CS departments at Auckland, Wellington and Otago were not established until the 1980s, outside our study period. In Auckland at least, this reluctance was caused partly by considerations about duplication of effort, but also by assertions that Computer Science was not a "proper" science - one view was even that computing was merely a branch of physics [23]. At Otago, Brian Cox was appointed as "Lecturer in Charge of the Computer Centre" in 1964, even though the first computer, an IBM 360/30, did not arrive until 1966. Cox had used the EDSAC 2 in Cambridge as a physics graduate student, trained by Maurice Wilkes and David Barron. Although computing courses were taught from 1966, and Computer Science papers from 1972, Cox had to wait until 1984 to become the founding Professor of Computer Science [15].

LOCAL CONTRIBUTIONS

TBD: Local contributions to the technology [Mano]

In 1966, Bruce Moon (then at the University of Canterbury) published the first local programming textbook [33].

In 1968, Progeni, claimed to be the first software house in New Zealand, was founded by Perce Harpham (1932-2xxx) [27].

[Brian's anecdote 1:] In 1974, with increasing demand for interactive computing, Massey University decided that Burroughs block-mode terminals were too expensive. Instead, the Computer Unit bought a DEC PDP 11/10 to act as a terminal concentrator for inexpensive DEC VT05 terminals. Lacking software, the only answer was DIY. Brian Carpenter converted CERN's compiler for the PL-11 programming language to run on the Burroughs mainframe, and Ted Drawneek wrote the necessary software.

[Brian's anecdote 2:] In 1975, teams at Massey University and Victoria University of Wellington launched a project called Kiwinet, whose initial goal was resource sharing between the B6700s at the two universities. Despite establishing connectivity, there was no software ecosystem and the project was mainly a learning experience, as discussed in Chapter 2 of [34]. Internet connectivity only came to New Zealand in 1989.

ECONOMIC IMPACT

In 1960, the New Zealand economy was very tightly coupled to the UK, despite being literally a world away. The New Zealand currency was held at par with the British pound, and the UK accounted for 53% of exports and 44% of imports. Moreover, 90% of the total value of exports was derived from agricultural produce [1]. However, as early as 1961, Britain announced its interest in joining the European Economic Community (EEC), and this was correctly interpreted in New Zealand as a serious warning that access to the UK market would sooner or later become difficult.

In fact, diversification had already started, with meat exports going to 60 different countries. There was a successful effort at further diversification of both exports and imports. For example, tariff reductions were negotiated with Japan, and a free trade agreement with Australia was signed in 1965 [35]. Although New Zealand's access to UK markets was eased by a so-called Luxembourg agreement in 1971, when Britain eventually joined the EEC in 1973, New Zealand had already become a completely independent trading nation, living on its wits, as it remains today. As noted above, IBM and other American computer companies took advantage of this opportunity. By 1975 exports to the UK were down to 22% of the total and imports to 19% [13].

Thus the growth of computing in New Zealand took place against a background of major economic upheaval. This makes it difficult to quantify the economic effects of computing. It is safe to assume that the companies and other organizations that increasingly used computers saw a financial advantage in doing so. As noted above, project planning (PERT) was a widespread application, which suggests that companies saw much more to computing than savings on clerical staff. Writing of government computers in 1985, A. C. Shailes [37] said "The use of computers has released staff from mundane administrative support tasks and allowed them to become directly involved in achievement of the government's main objectives. It is clear that direct financial savings from computers have recovered the overall cost of all computer resources many times over." On the banking front, Ian Archibald [19] wrote of Databank that "co-operation in basic processing systems has freed up staff to intensify competition in developing and marketing the banking products." Quantifying these effects in millions of dollars seems impossible, but we conclude that the kind of competitive and flexible economy that New Zealand needed to survive as a trading nation by 1975 would have been essentially impossible had it lagged in computing.

SOCIAL IMPACT

TBD: Check for completeness etc

Perhaps it is no surprise that lawyers were among the first to consider the social impact of computers. A first legal analysis was published in 1971 by Francis M. Auburn [20], in which consideration of privacy was the major theme. The author observed that "there is a fear of computers and their operators among the general public" and quoted a rather naive counter-argument that "invasions of privacy are the result of misuses of the data retrieved and not a function of the storage medium" (i.e., blame the users, not the technology). With hundreds of thousands of banking transactions per day already taking place, and with a national police computer and computerized hospital records in the offing, the risks of accidental or intentional privacy breaches already seemed very real, more than 50 years ago. The legal situation was murky: "If there is a general right of privacy we are still no nearer to a definition of the circumstances in which it is applicable to computers." All sorts of data raised thorny privacy issues once computerized, such as medical records, census returns, and credit history. The issue of a unique national identity number to tie all

FIGURE 7. Wanganui Computer Centre, 1975. Courtesy Fletcher Archives, NZ.

FIGURE 8. BNZ Ledger Room, 1950s. Courtesy BNZ Museum.

records together was also of concern, with inevitable references to George Orwell's novel *1984*. Auburn commented that "The most rigorous presently known technological safeguards would be quite insufficient to prevent abuse." Auburn's conclusion was that a "comprehensive review of the present and future impact of electronic data processing on society" was needed, to be followed by legislation.

Auburn was not alone in raising these concerns. For example, also in 1971, A. L. C. Humphreys of ICL, the Christchurch *Press*, and the New Zealand Computer Society were all concerned about privacy. Certain lawyers even drafted a proposed law [6] and a Bill was introduced to Parliament in 1975 [11] but never made it into law. New Zealand's first Privacy Act was not passed until 1993, prior to which common law applied.

Despite this legal vacuum, plans for a national police database were made from 1972. The system, officially the National Law Enforcement Data Base, often called the Wanganui Computer since it was installed in the town now known as Whanganui, was operational by 1976 (Figure 7). This led some years later to protest actions. However, it seems that during our study period, despite the concerns of some lawyers, there was little public perception of computers as a societal threat prior to the police database.

Apart from privacy issues, it is hard to separate the impact of computers from other factors that changed society in the 1960s and early 1970s, as the first generation born after World War II grew to maturity. Colin Beardon [21] discussed the computerization of "women's work" and its possible impact on unemployment. Certainly, skilled women who operated ledger machines and comptometers found themselves out of a job as computers took over (Figure 8). Some of them no doubt moved into computer operations. However, many of the replacement jobs would have been less skilled and lower paid data entry jobs. Beardon noted that the fraction of the "unemployed who are female" increased from 10.9% in 1961 to 18.0% in 1971 and 34.9% in 1975, but is impossible to know how much of this change was caused by computers. For example, in retail banking, according to anecdotal evidence, the technological change from manual to computerized operations changed people's jobs, but did not destroy them [31]. The loss of typing and secretarial jobs came a decade later, with the advent of personal computers. Throughout our period of study, computers were mainly hidden from public view except for people directly involved, and had relatively little social impact compared to other modern technology such as television. Apart from the job market, where computers had a direct effect on some working lives, they were not even a talking point for most people.

CONCLUSION

TBD

ACKNOWLEDGMENTS

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